Loops and Stopping

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One connected while-loop progress record.

Trace all checks, compare zero/one/several updates and repair equality-only stopping before independent transfer.

Keep one habit: Keep the condition, update, tested values, final state and actual evidence together.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from **Trace Table**s. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Decompose a Goal. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Trace Tables

Decompose a Goal

MAIN LESSON · SECTION 12

W1 · Trace every while-condition check

> Use the supplied complete script, including the final false test.

Current while-loop task and answer

- For count=12, while count>3 subtract 4, list every value tested by the condition, including the final false test. How many updates occur, and what is printed?

```
count = 12
while count > 3:
    count = count - 4
print(count)
```

> **My while-loop reasoning and evidence**

▼ **Earlier for-loop task and saved response**

List the values supplied by range(2, 5) and explain why 5 is absent. How many times does its body run?

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Run zero, one and several updates

> Keep expectations separate from actual fresh runs.

Current while-loop task and answer

- Using the complete while script, predict fresh starts 3,4,12. Record tested values, truth results, updates and actual **output** separately. Explain the zero-update case.

My while-loop reasoning and evidence

▼ **Earlier for-loop task and saved response**

Copy the worked script. Change the range to range(2, 5). Record turn, total before and total after each **iteration**. Predict the single **output**, then run and compare.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Repair equality-only stopping

> Use a finite paper prefix rather than executing the unbounded variant.

Current while-loop task and answer

- Replace count>3 with count!=3, starting at 12 and subtracting 4. Trace only five updates on paper, explain why equality is missed and restore the appropriate condition.

My while-loop reasoning and evidence

▼ Earlier for-loop task and saved response

For the faulty script with range(1, 4), record total after each addition. Explain the wrong final **output** and remove only the faulty reset.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Expose a hidden missing update

> An initially false condition cannot test progress inside the body.

Current while-loop task and answer

- A loop with no update is tested only with count=3 and stops. Does that establish termination from count=12? Explain and choose starts giving zero, one and several updates in the correct script.

My while-loop reasoning and evidence

▼ **Earlier for-loop task and saved response**

In your counter script, range(-2) performs no **iteration**s. Does returning the initial 5 make negative rounds valid? Explain what the script checks and what it assumes.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Changed independent case

> Keep one first attempt before hints or feedback.

Read and explain independently

Start count at 1. While count < 8, add 3. Trace the condition and values. What happens if the condition becomes count != 8?

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Changed independent case

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Loop through every candidate book, **update** its score, then stop at the end of the list.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Trace before running for **Loop**s and **trace table**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for **Loop**s and **trace table**s. Proposed explanation: “The stop value is included”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Trace count=0; while count<4: count=count+2. List values tested by the condition and final count. Would replacing <4 with !=5 terminate?

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Start n=0; while n<3, add 1 and record n.

List records and stopping condition.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

Start at count 0. While count is less than 3, add 1 and display count. List the displays and the stopping value.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 5 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Reference · Original target response space

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Trace count=0; while count<4: count=count+2. List values tested by the condition and final count. Would replacing <4 with !=5 terminate?

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Extra practice · Previous for-loop counters

REFERENCE · SECTION 15

Extra practice · Previous for-loop counters

> This earlier range task is review; W4 is the selected increasing while-loop assessment.

Try the prior range task

- A box starts with 5 counters. Each round adds 2 counters. Use rounds = 3, an initialized total and for turn in range(rounds). Print the final total once after the **loop**.

  Trace the totals and predict fresh runs with rounds 0, 1 and 3. Assume rounds is a nonnegative integer.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

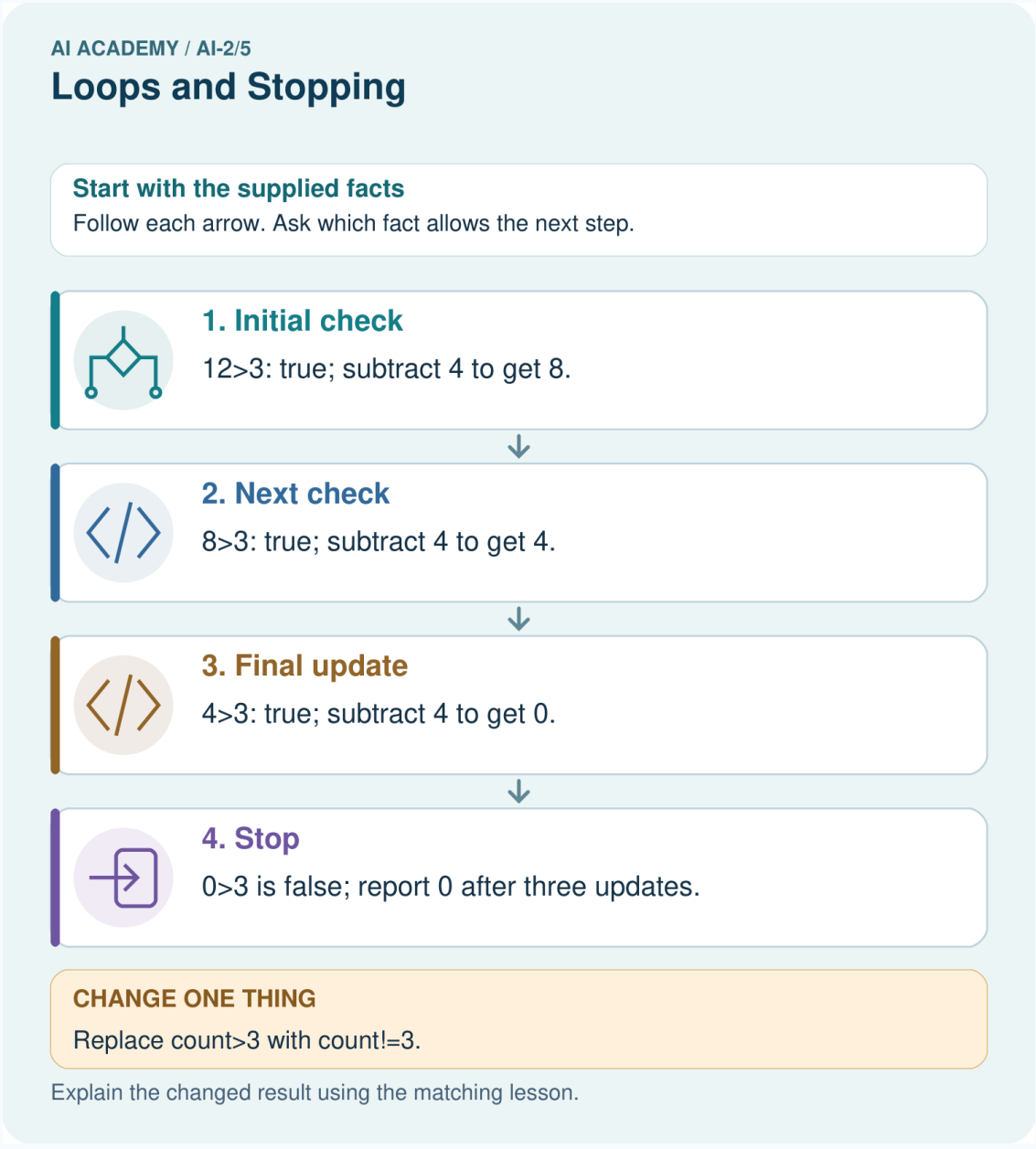
> Keep the explanation and the support you used.

Use this when helpful

A fictional counter starts at 12 and decreases while above a boundary.

Python: count=12; while count>3: count=count-4; print after the **loop**.

The condition is checked before each subtraction.



Follow the labelled picture with the matching explanation. Open the vector diagram

Why is report 0 correct for the original **loop**, and why is equality-only stopping risky here?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Use progress and stopping arguments for any repeated recommender **process**; test **input**s that require zero, one and several **update**s.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Transparent Book Recommender

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use progress and stopping arguments for any repeated recommender **process**; test **input**s that require zero, one and several **update**s.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional counter starts at 12 and decreases while above a boundary.
- Which labelled step matches this detail? 0>3 is false; report 0 after three **update**s.
- 1. Next check
- 2. Final **update**
- 3. Stop

- 4. Initial check

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Stop 0>3 is false; report 0 after three **update**s. The condition does not promise the counter ends exactly at 3. Subtractions can cross the boundary.

The count strictly decreases, so the below-or-equal stopping region is eventually reached from this start. A condition requiring exact equality can be unsafe when repeated **update**s skip that value.

Changed-case question

- Changed-case practice: Replace count>3 with count!=3. Which conclusion is supported by the supplied facts?

- 1. We can decide the result without examining the changed **input** or the procedure.

- 2. From 12, the **loop** never reaches 3 and does not terminate in this toy model.

- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: From 12, the **loop** never reaches 3 and does not terminate in this toy model. 12,8,4,0,−4,... are multiples of 4; none equals 3.

Repair and save your work

Review **Trace Table**s (ai-2/4). Goal: Trace variable values before and after each **loop iteration**. Short practice: Move print inside the original **loop**. What sequence is displayed?.

Return to this week and explain what you changed.

Review Decompose a Goal (ai-2/2). Goal: Distinguish a function definition from a function call. Short practice: What happens visibly if the caller stores needed but omits print(needed)?.

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

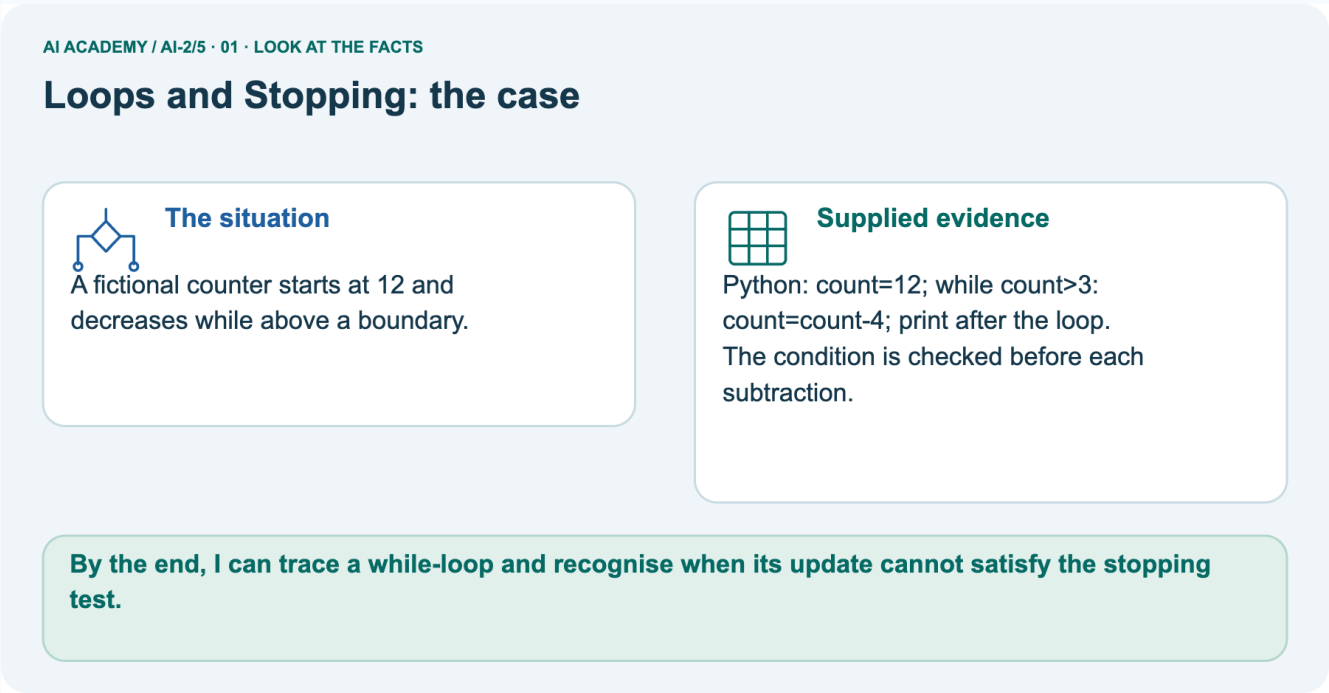
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Loops and Stopping: the case



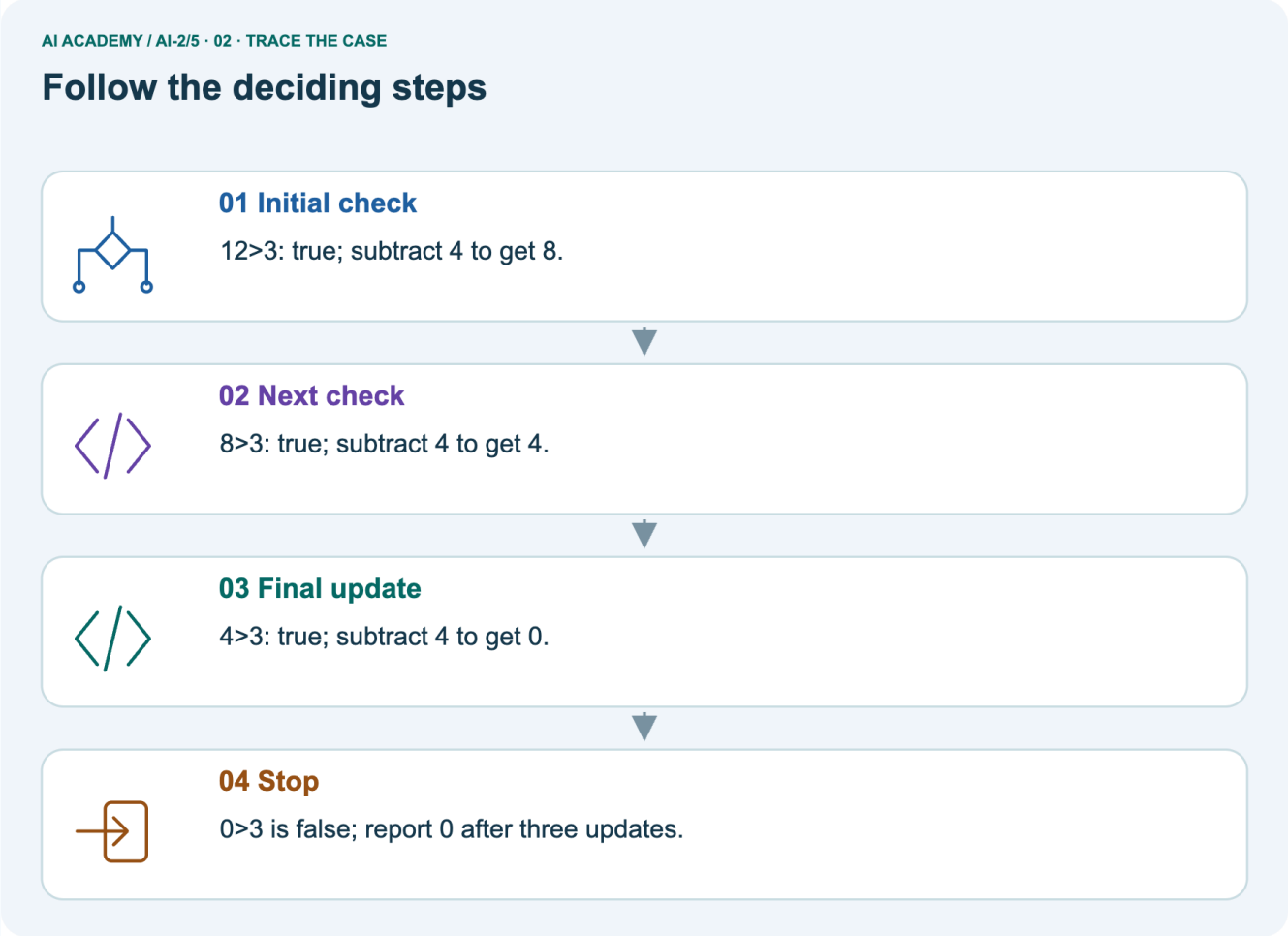
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional counter starts at 12 and decreases while above a boundary. Supplied facts: Python: count=12; while count>3: count=count-4; print after the **loop**. The condition is checked before each subtraction.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Initial check: 12>3: true; subtract 4 to get 8. → Next check: 8>3: true; subtract 4 to get 4. → Final **update**: 4>3: true; subtract 4 to get 0. → Stop: 0>3 is false; report 0 after three **update**s.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. State after each addition

AI ACADEMY / AI-2/5 · 03 · INSPECT THE LESSON MODEL

State after each addition

turn	total before	total after
1	0	1
2	1	3
3	3	6

Initialize total to 0 before the loop. The unindented print runs once afterward.

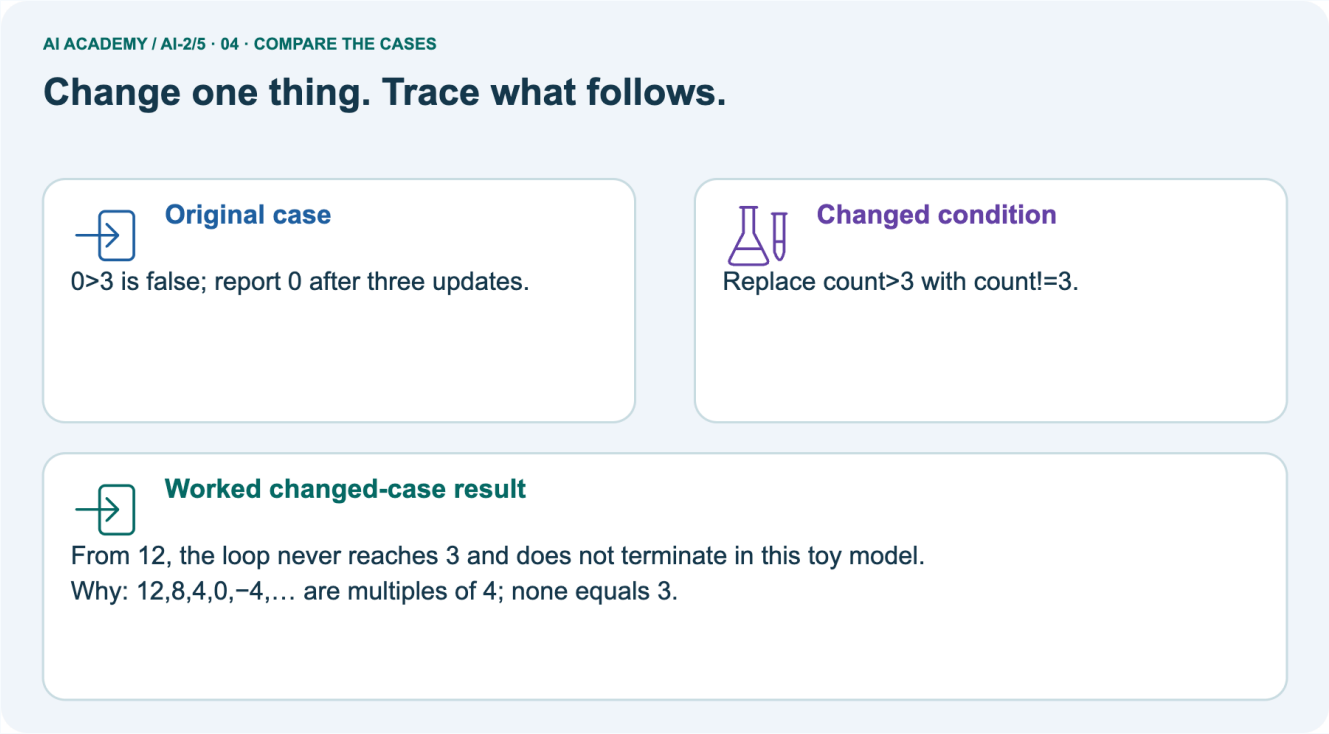
Follow the labelled picture with the matching explanation. Open the vector diagram

State after each addition. 1 | 0 | 1. 2 | 1 | 3. 3 | 3 | 6

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

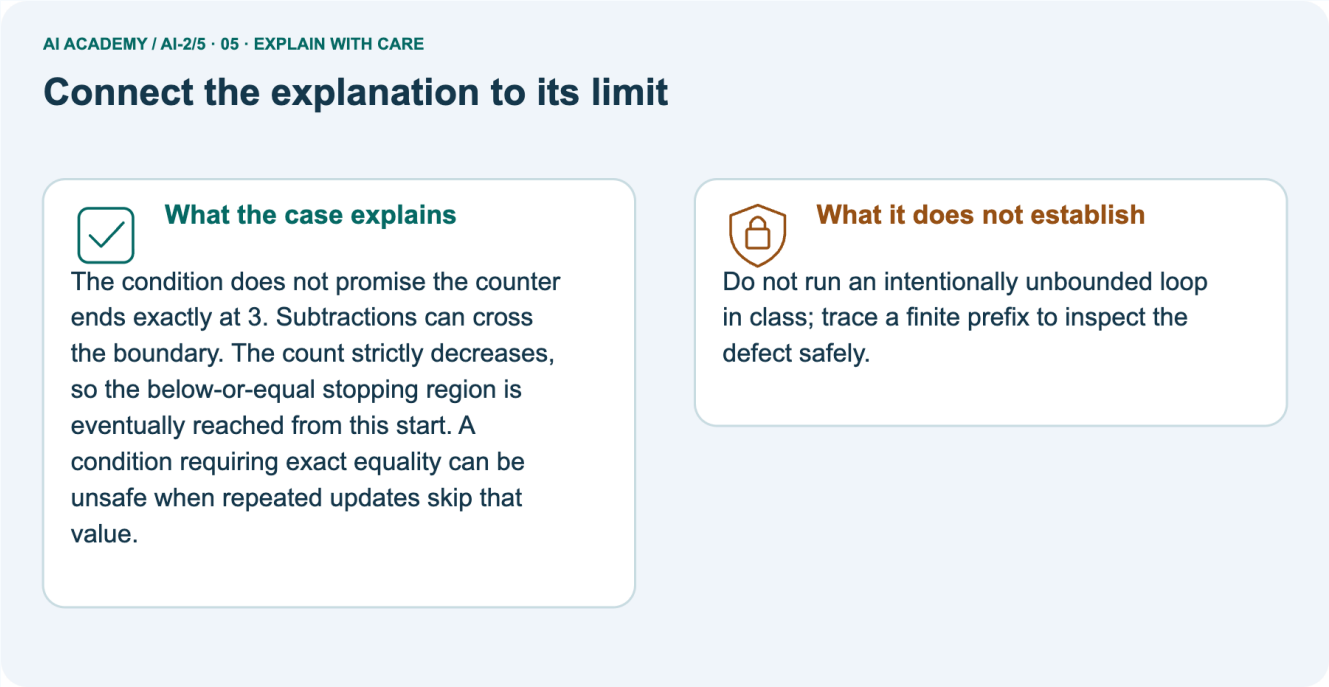
Original case: 0>3 is false; report 0 after three **update**s. Changed condition: Replace count>3 with count!=3. Expected result: From 12, the **loop** never reaches 3 and does not terminate in this toy model.

Reason: 12,8,4,0,−4,... are multiples of 4; none equals 3.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

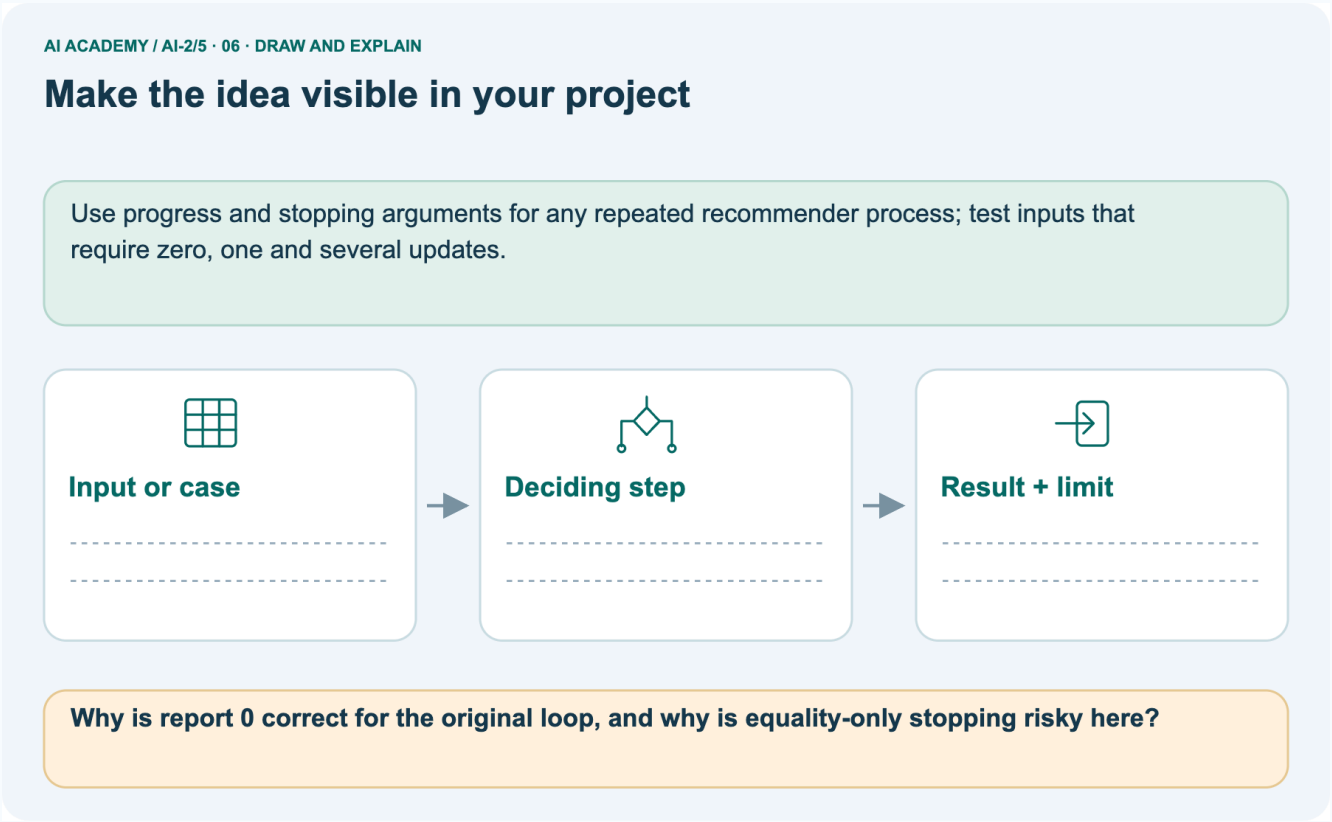
The condition does not promise the counter ends exactly at 3. Subtractions can cross the boundary. The count strictly decreases, so the below-or-equal stopping region is eventually reached from this start.

A condition requiring exact equality can be unsafe when repeated **update**s skip that value. Limit: Do not run an intentionally unbounded **loop** in class; trace a finite prefix to inspect the defect safely.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Use progress and stopping arguments for any repeated recommender **process**; test **input**s that require zero, one and several **update**s. Draw your **input**, deciding step, result and limit.

Supported question: Why is report 0 correct for the original **loop**, and why is equality-only stopping risky here?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Loops and Stopping: the case

02. Follow the deciding steps

03. State after each addition

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.